IST 691: Research Methodology and Scientific Writing Skills

Introduction

Denis L. Nkweteyim

Outline

- What is Science?
- What is research?
- Methods of acquiring knowledge
- The scientific approach
- Objectives of science
- Basic assumptions underlying science
- Method vs technique

What is Science

- Some definitions
  - A method or logic of inquiry
  - A systematic endeavor that builds and organizes knowledge in the form of testable explanations and predictions about the universe (Wikipedia)
  - The systematic study of the structure and behaviour of the physical and natural world through observation, experimentation, and the testing of theories against the evidence obtained

What is Computer Science?

- The systematic study of computing systems and computation
- Computer science knowledge includes among others, theories for understanding
  - Computing systems and methods
  - Design methodology
  - Algorithms, and tools
  - Methods for the testing of concepts
  - Methods of analysis and verification; and
  - Knowledge representation and implementation

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What is research?

- What is research?
  - Human activity based on intellectual investigation and aimed at discovering, interpreting, and revising human knowledge on different aspects of the world
- Normally based on the Scientific Method
  - Research can use the scientific method, but need not do so (Wikipedia)

Why do Research?

- Research exposes the world
- Investigate for the purpose of
  - Discovering something new
  - Interpreting something not understood
  - Revising something wrong or incomplete

What is Computer Science Research?

- Systematic investigation and study of computer-related topics to advance knowledge, develop new technologies, and solve complex problems
- Involves the application of scientific methods and techniques to explore various aspects of computer science
  - Algorithms, programming languages, artificial intelligence, databases, computer systems, networking, software engineering, big data analytics, machine learning, natural language processing, digital image processing, Internet of Things (IoT), cloud computing, computer vision, etc.
- Objective is to expand the understanding of fundamental principles, discover new algorithms, invent innovative approaches, and contribute to the development of cutting-edge technologies
- It often involves formulating hypotheses, conducting experiments or simulations, analyzing data, and/or drawing conclusions based on empirical evidence

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Methods of Acquiring Knowledge

- At least six ways to obtain information about a given phenomenon or situation, but only one of them is the scientific method
- We first consider some of the unscientific approaches to acquiring knowledge
- Note
  - We are not saying that these unscientific approaches have no place in science
  - While they do not contribute to the accumulation of scientific knowledge, they are used in the process of ***implementing*** the scientific method
- The five *unscientific* approaches follow

Methods of Acquiring Knowledge - Tenacity

- Tenacity
  - Holding on to some habit (more like superstition)
  - Continue believing in something we have always believed
- Problems with tenacity
  - Knowledge that is acquired through mere exposure may be inaccurate
  - Tenacity does not provide a mechanism for correcting erroneous superstitions and habits in the face of evidence to the contrary

Methods of Acquiring Knowledge - Intuition

- Intuition
  - When we use our intuition, we are relying on our guts, our emotions, and/or our instincts to guide us, rather than examining facts or using rational thought
  - Problem
    - Intuitions can be wrong because they are driven by cognitive and motivational biases rather than logical reasoning or scientific evidence
  - Positive
    - Weighing alternatives and [logical] thinking of all the different possibilities can be paralyzing for some people and sometimes decisions based on intuition are actually superior to those based on analysis

Methods of Acquiring Knowledge - Authority

- Authority
  - One of the most common methods of acquiring knowledge
  - Method involves accepting new ideas because some authority figure states that they are true
  - Problem
    - The authority we are relying on may be wrong
    - Nevertheless, much of the information we acquire is through authority because we don't have time to question and independently research every piece of knowledge we learn through authority
    - But we can also learn to evaluate the credentials of authority figures, to evaluate [using the scientific method] the methods they used to arrive at their conclusions, and evaluate whether they have any reasons to mislead us

Methods of Acquiring Knowledge - Rationalism

- Rationalism
  - Involves using logic and reasoning to acquire new knowledge
  - Premises are stated and logical rules are followed to arrive at sound conclusions
  - Problem with the method
    - If the premises are wrong or there is an error in logic then the conclusion will not be valid
    - Also, unless formally trained in the rules of logic it is easy to make an error

Methods of Acquiring Knowledge - Empiricism

- Empiricism
  - Involves acquiring knowledge through observation and experience
    - Probably the most obvious approach that is used in science, as science is based on observation
  - Problem with empiricism
    - Observations and experience may be wrong

Methods of Acquiring Knowledge - Empiricism

- Examples of problems with empiricism
  a) For centuries people believed the world is flat because it appears to be flat
  b) From psychology, we learn of several visual illusions that trick our senses that we are perceiving something different from what we are actually perceiving
  c) Experiments on investigating satanic messages when some recordings are played backwards
     1) One researcher played records backwards and asked people to hear for themselves the satanic messages that appeared on the records
     2) Other researchers repeated the experiment but subjects were simply asked to identify what they heard when the records were played backwards

Empiricism (observations) was used in each case; however, in the first case, experimenter proposed to the subjects that the the source of the message was satanic, thereby generating expectations on the type of message that might be on the records

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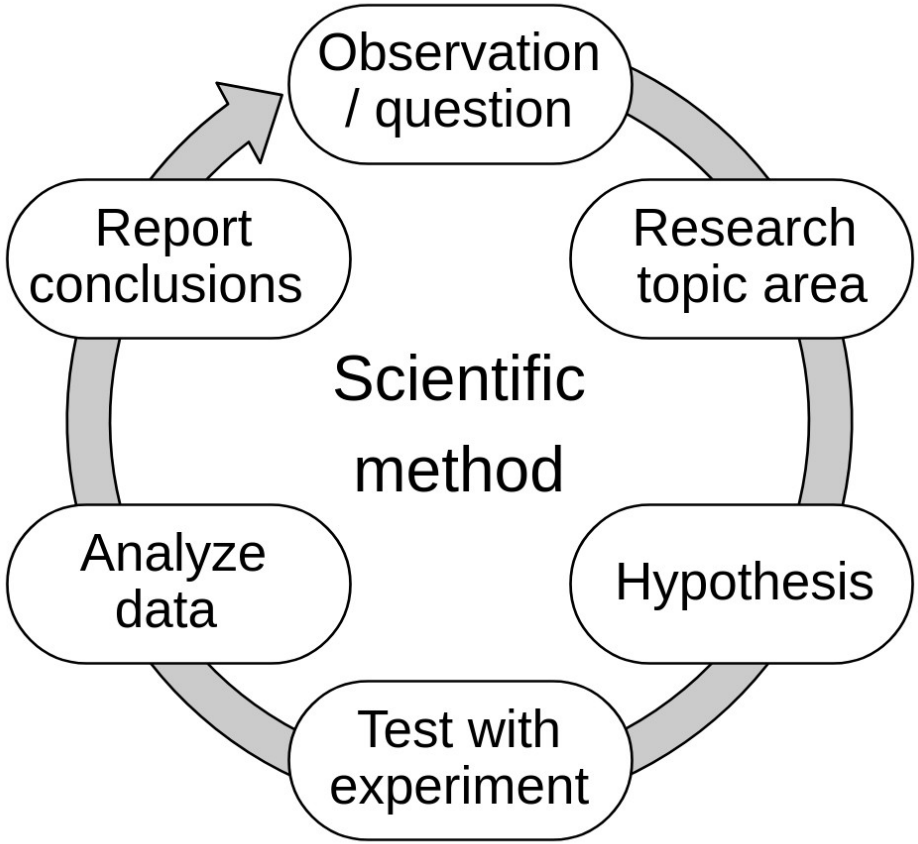
The Scientific Method

- Empiricism is vital in science
- But the collection of data must be through the use of the scientific method, not through personal experience of an event

The Scientific Method

- Most likely of all of the methods to produce valid knowledge
- Process of systematically collecting and evaluating evidence to test ideas and answer questions
- While scientists may use intuition, authority, rationalism, and empiricism to generate new ideas they don't stop there. Scientists go a step further by using systematic empiricism to make careful observations under various controlled conditions in order to test their ideas and they use rationalism to arrive at valid conclusions
- Drawbacks
  - Not always feasible to use the scientific method
    - May require too much time and resources
    - Cannot be used to answer all questions - can only be used to address empirical questions

What is the Scientific Method?



What is the Scientific Method?

- The scientific method
  - A process for experimentation that is used to explore observations and answer questions
- Not all scientists follow the scientific method exactly?
  - Some areas of science can be more easily tested than others
  - Hence, when direct experimentation is not possible, scientists modify the scientific method
  - Even when modified though, the goal (and many of the steps) remains the same
    - i.e., to discover cause and effect relationships by asking questions, carefully gathering and examining the evidence, and seeing if all the available information can be combined into a logical answer
  - New information or thinking might also cause a scientist to back up and repeat steps at any point during the process

Steps in the Scientific Method

- Identify the problem and formulate a hypothesis
- Design the experiment
- Conduct the experiment
- Test the hypothesis
- Communicate the research results

Characteristics of the Scientific Approach

- Control
  - Most important element in the scientific method
    - Control refers to elimination of extraneous variables that could affect the observations that are made
    - Enables scientists to identify the causes of their observations
- Operational definition
  - The definition of terms by the steps or operations used to attain or measure them
  - This eliminates confusion in communication
  - Example
    - Hunger causes one to perceive food-related objects selectively
    - Must define what we mean by hunger
      - Hunger refers to being starved (ambiguous)
      - Hunger refers to eight hours of food deprivation (clear definition)
- Replication
  - Reproduction of the results obtained from a study
  - Observations made must be replicable
    - i.e., the same results must be found if the study is repeated

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Objectives of Science

- Ultimate objective of science
  - To understand the world in which we live
- But understanding the world means different things to different people
- Science demands detailed examination of a phenomenon
  - Only when a phenomenon is accurateley described and explained – and therefore predictable and in most cases capable of being controlled – will science say that it is understood

Objectives of Science

- Description
  - Portray the phenomenom accurately, identify the variables that exist, and the degree to which they exist
- Explanation
  - Determination of the cause of a given phenomenom
- Prediction
  - Ability to anticipate the occurrence of an event
- Control
  - Manipulation of the conditions that determine a phenomenom

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Basic Assumptions Underlying Science

- Two basic assumptions that enable scientists to have confidence in the capacity of scientific inquiry to achieve a solution to questions and problems
  - Uniformity in Nature
  - Determinism

Uniformity in Nature

- Scientists believe that there is uniformity in nature
  - Science is a search for order, for uniformities, for lawful relations among events in nature
  - Without uniformity
    - There could be no understanding, explanation, or knowledge about nature
    - We could not develop theories, laws, or facts

Determinism

- Determinism
  - The belief that behaviour is caused by specific events
  - Determinism is implicit in the assumption of the uniformity in nature
- In the effort to uncover the uniform laws of behaviour, we attempt to find the variables that are linked together
  - Through experiments to try to establish the determinants of events

Basic Assumptions Underlying Science

- Several axioms to enable the scientist to uncover the deterministic relationships inherent in the uniformity of nature
  - Reality
  - Rationality
  - Regularity
  - Discoverability

Axioms Related to Scientific Inquiries

- Reality in nature
  - The assumption that the things we see, hear, feel, and taste are real and have substance
- Rationality
  - The assumption that there is a rational basis for the events that occur in nature and they can be understood through the use of logical thinking
- Regularity
  - The assumption that events in nature follow the same laws and occur the same way at all times and places
- Discoverability
  - The assumption that it is possible to discover the uniformities that exist in nature

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Method vs Technique

- Important to distinguish the scientific method from the technique used
- All sciences are characterized by a common logic of inquiry
- However, different techniques are used to apply the scientific technique depending on subject matter, nature of the problem, and the stage of the inquiry
- Examples
  - Astronomer uses a telescope in making observations
  - A biologist may use a microscope
  - A small group researcher may use a one-way mirror to unobtrusively observe participants' interaction